

Implementation Plan: Custom MLLM-Geo-AI Urban Classification

Project Overview

Technical implementation strategy for Urban Classification. Goal: Classify geographic units into Residential, Commercial, or Industrial zones using POI data and Custom MLLM-Geo-AI.

Phase I: Spatial Data Engineering & Gridding

1. Data Loading: Import 'project.csv' and filter out points outside the study area.
2. Coordinate Projection: Project longitude (X) and latitude (Y) into a metric system (EPSG:3857).
3. Grid Generation: Create a spatial mesh of 500m x 500m cells.
4. Spatial Join: Assign each POI to its respective Grid ID.

Phase II: Feature Engineering (The MLLM Input)

1. Numerical Features: Calculate frequency counts of 'place type' for each cell.
2. Textual Aggregation: For each cell, concatenate the 'text_des' of all internal POIs into a single paragraph.
3. Text Vectorization: Use a Transformer model to convert paragraphs into numerical embeddings.

Phase III: Model Architecture & Training

1. Model Selection: Build a classifier (e.g., Random Forest or Neural Net) using Text Embeddings + Numerical Features.
2. Labeling (Ground Truth):
 - Residential: Education, Health, Religion, Leisure.
 - Commercial: Food, Shop, Finance, Tourism.
 - Industrial: Industrial, Transport.
3. Training: Train on 70% of the grid data and validate on 30%.

Phase IV: Visualization & Final Outputs

1. Prediction: Generate labels for all grid cells.
2. Mapping: Create the final Urban Classification Map with color codes:
 - Yellow: Residential
 - Red: Commercial
 - Purple: Industrial
3. Final Export: Output high-resolution PDF map and GeoJSON result.

Technical Stack

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- Language: Python
- Libraries: Pandas, GeoPandas, Scikit-learn, Sentence-Transformers, Matplotlib.
- Data: project.csv (Input), Final_Urban_Map.pdf (Output).